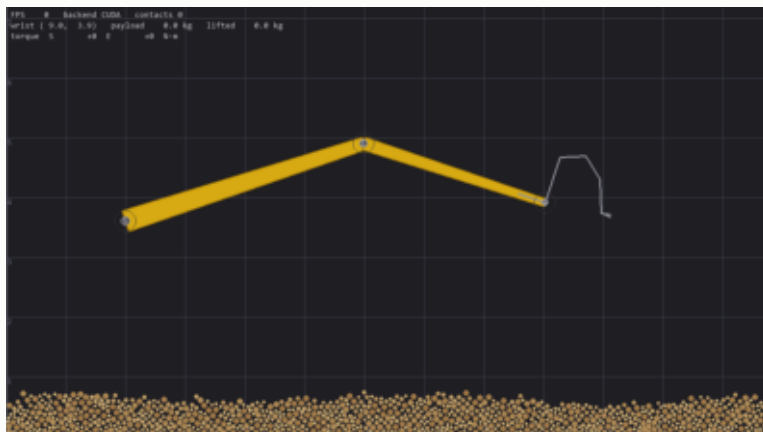
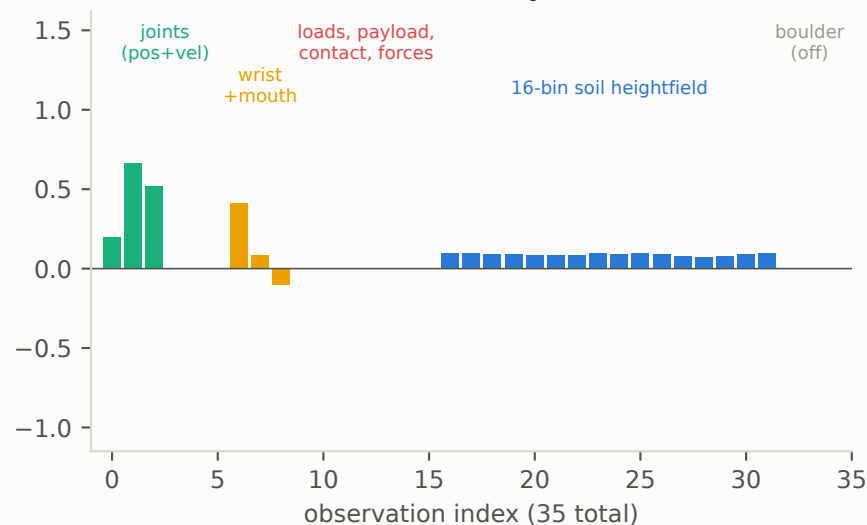


One state, four representations: from physics to the transport map

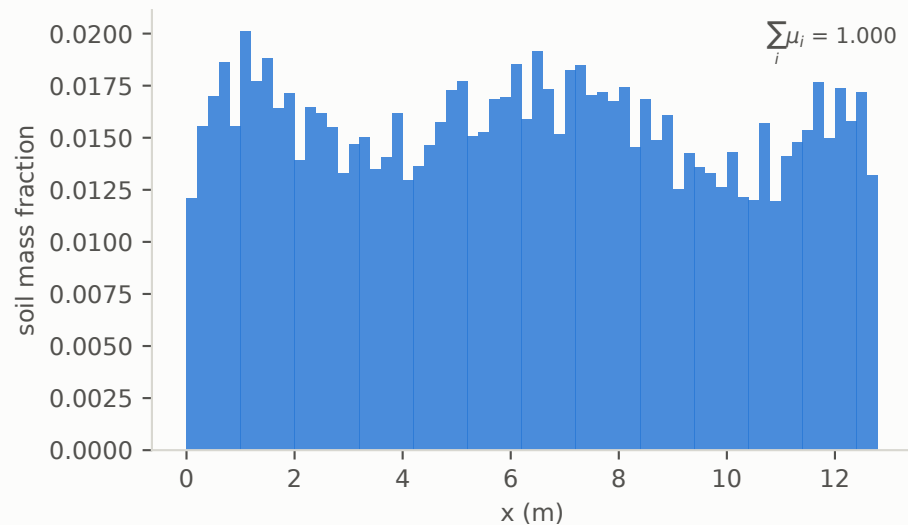
(a) The world: ~1200 simulated grains + 3-joint arm



(b) What the POLICY sees: body + coarse soil (35-dim)



(c) What the OBJECTIVE sees: $\mu = 64$ -bin soil mass distribution -- zero robot information, sums to 1



(d) The learned map: $\phi(\mu) \in \mathbb{R}^4$ (first 2 dims) skills travel directions; distance $\approx$ physical transport

